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African Interlace as Dynamic Grids: New Heritage Algorithms for Diaspora Design Ecologies

Ron Eglash  and Audrey G. Bennett 

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ABSTRACT African interlaces are continuous “over-under” paths that can be found in designs throughout the African continent. They include lusona drawings, Orisha beadwork, and related designs from hair braiding, textiles, and other media. But there has been little research examining the common cultural themes expressed, or the common elements of their shape constructions. We begin with an ethnocomputing methodology to show how their visualization of “circulating vital energy” is expressed by their formal properties as “dynamic grids.” In the second part of the paper, we examine their relationship to diverse Black interlace designs in the Americas: cornrow hairstyling, Haitian vèvè, Brazilian capoeira, and other forms show dynamic grids as “heritage algorithms.” The final part of the paper looks at how contemporary Black designers have utilized these cultural and formal properties as heritage algorithms, highlighting

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their contribution to liberatory practices in what we are calling “diaspora design ecologies.”

KEYWORDS: heritage algorithms, dynamic grids, lusona, African interlace, Black designers, diaspora design ecologies

Introduction

Black cultural creations have been instrumental to histories of resistance and survival. They are increasingly important forms of “cultural design capital” that empower disenfranchised communities (Bennett 2003; Berry et al. 2022). Blending historical and contemporary elements of Black cultural practices with AI, architecture, art, data sovereignty, food systems, human computer interaction (HCI), media design, and other pathways can contribute to the emerging diaspora design ecologies. These networks of practitioners, theorists, and activists have developed new innovations for embedding anti-racist and liberatory practices, humanizing technology, democratizing socio-economic systems, and other transformations that embody generative justice. Eglash (2016) defines this as:

The universal right to generate unalienated value and directly participate in its benefits; the rights of value generators to create their own conditions of production; and the rights of communities of value generation to nurture self-sustaining paths for its circulation.

This paper introduces a new form of Black design capital with generative implications: the African interlace tradition. As “dynamic grids,” African interlace mediates between fixed measures of static shapes and relationships of connectivity that can flourish in flexible and dynamic spaces. Interlaces are thus both an “eco-epistemology,”¹ reflecting African heritage practices such as relational economies and ubuntu ethics,² and a powerful new asset for Black design practices in both physical and virtual domains.

Methodology

Standard accounts of comparative knowledge systems assume a “ladder of knowledge” in which Indigenous societies might have detailed facts about plants and ecosystems, but not high level mathematical or scientific concepts. Here we utilize the “bush” knowledge structure from ethnocomputing, which maintains that understandings that come late in some societies might come earlier in others; each cultural branch could be a different sequence. This was proposed in the first work on fractal geometry as an Indigenous knowledge system in Africa (Eglash 1999), and its method is described below.

Ethnocomputing begins with observations of design patterns. But rather than impose the researchers’ understandings, it uses ethnographic accounts of the cultural meanings and techniques to

discover the practitioner's own conceptions of the design construction, i.e. the "heritage algorithm" (Bennett 2016). For example, Navajo rug weavers use a pattern such as "up one over one" to create a diagonal. But that is not a 45 degree angle, because the vertical warp thread is thicker than the horizontal weft thread. Babbitt, Lyles, and Eglash (2012) found that weavers had developed their own algorithms that did not match those created on a Cartesian grid, and this had consequences for their own priorities around aesthetic, spiritual, and economic meanings and practices. In the work below, we take a similar approach, surveying not simply the geometric properties of the interlace patterns in Africa, but the relationship between their modes of construction and the underlying cultural meanings and motivations.

One important consequence of the ethnocomputing method is that it makes the "heritage algorithm" available for contemporary uses of the Indigenous design. For example, African fractals are now an explicit part of contemporary architecture in Ethiopia, utilized in STEM education software, Black arts, dance, literature, and other domains (Eglash et al. 2021, 2024). We refer to this kind of distribution of culture-based practices for design as "diaspora design ecologies." Investigating how the heritage algorithms of African interlace might be a basis for its own diaspora design ecology, similar to the ways that African fractals did so, was an important research question for this paper, and we will return to that in the final section.

Interlace Patterns in Africa

Introducing African Interlaces through Lusona, an Algorithmic Tradition from Central Africa

Lusona sand drawings are a traditional practice by the Chokwe, Lwena, Luchazi, Lunda, Lwimbi, Songo, and Ovimbundu, and others in Angola, northwestern Zambia, and the Democratic Republic of the Congo. In this tradition, a continuous line is traced around a grid of spaced dots. You cannot lift your stylus from the surface or trace over the same line twice.³ Most lusona end where they began, a property called "unicursal" in topology. A simple example of a unicursal lusona is illustrated in the top row of Figure 1.

As it is drawn, the artist tells a story about the figure, often describing movements through a story such as "leopard chasing the stork." You can see in Figure 1 that the trace in sand creates an over versus under effect, such that the history of the trace is made visible: hence our term "dynamic." In some cases, the graphic algorithm (set of drawing steps) can be expanded into increasingly complex versions, creating a "family" of algorithms. The top row of Figure 1 shows a grid of dots, one row with two columns (1 by 2). The next two rows show how that family expands into 2 by 3 and 3 by 4 versions. The potential for expansion (any grid of $n + 1$ by $n + 2$, starting with $n = 0$) is another aspect of this dynamism.

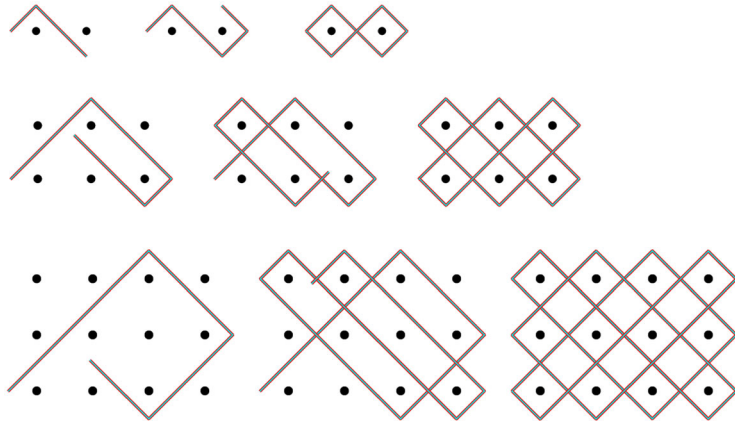


Figure 1

Three levels of the same family of lusona algorithms. Each level demonstrates the algorithm for drawing a continuous “over/under” line without lifting the stylus. Image courtesy Audrey Bennett.

We encourage the reader to recreate the paths in [Figure 1](#) with pencil and paper. You will soon find that you can create even higher levels of the design, even though it was never taught to you.⁴ When that happens, your head and hands have started to learn the “meta-algorithm” for producing all designs in this family. A meta-algorithm is a higher level algorithm for producing other algorithms. Such meta-algorithms are important to African traditions in several ways, and can be the liberating force for a diasporan design ecology.

Angolan traditional cultures, like many Indigenous systems in Africa, stressed the importance of a coming-of-age ritual. For example, in *Conversations with Ogotemmel* (Griaule 1965), a Dogon elder in Mali described how a lifetime of learning these knowledge layers, each creating the basis for the next level up, was required to start grasping their recursive cosmology’s unfolding.⁵ For youth, this begins in simple ways, as the knowledge that is needed to transition from child to teen, or teen to adult. In the Angolan case, this knowledge was conveyed in part by these story-telling lusona performances. By the time you are an adult you may have learned the meta-algorithm, but there is always a higher level. A lusona expert can be the maker of new metas, unfolding new stories, and so on, to the unfolding of the universe itself. By mapping the recursive series of lusona algorithms onto the recursion of their lives, the society establishes and renews its communal episteme in each generation (Eglash 1999).

This generative capability of the meta-algorithm is just one example of how lusona embody the concept of a “dynamic grid.” Kubik (1987) and Gerdes (1994) have documented additional ways in which Indigenous knowledge of the lusona’s interlace reflects a combination of fixed grids and dynamic, flexible paths. For example, Kubik points out that many lusona start with a rectangular grid, but

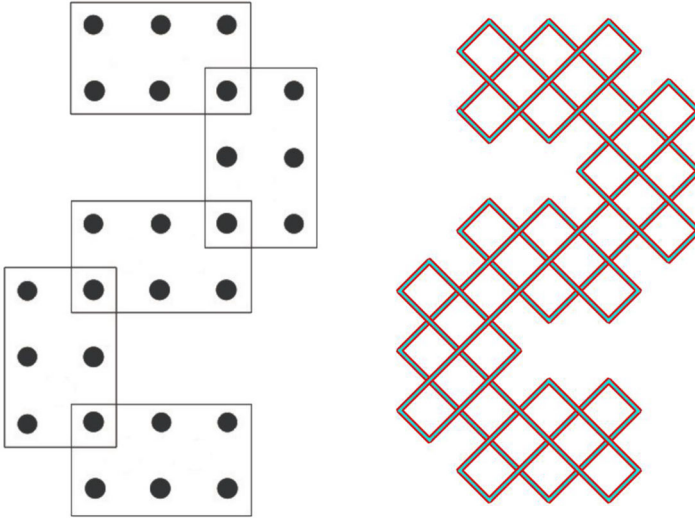


Figure 2

An overlapping grid concept at left, and the Indigenous conversion to a dynamic lusona path at right. Image by Audrey Bennett, based on a drawing in Kubic (1987).

then “chain together” with other rectangular grids (Figure 2). He notes that a similar process underscores the distinct polyrhythmic nature of African music. The musician is not bound to the beat in the fixed way that Western music tends to create or unmoored from structure as in avant garde music. Rather, traditional African music emerges from a framework that is *both guided and improvisational*. In similar ways, Henry Louis Gates (2014) characterizes Black communicative structures from both African and African American cultures as “repetition with revision.” Thus the social concept of a productive tension between guidance and innovation has symbolic representation in the productive tension between static grids and dynamic movement.

We tend to misinterpret lusona by focusing on the completed, static figures, as if they are simply screen icons to click on, each with a specific, singular meaning attached to it. But we would also be failing if we only understand it as movement and process. It is in the dynamic play between the two – the grid that guides the movement and the movement that transcends the grid – that the cultural epis-teme lives. Thus the math referred to in the West as “knot theory” has parallels to the lusona cultures’ metaphors of knots to describe the flow of “life force”:

The pervasive Luunda metaphors of the knot (lukat, puund, munuung), of knotting, tying, entwining, interjoining and connecting lie at the heart of many Luunda rituals to accrue life, vitality, fertility and well-being. (De Boeck 1993, 180)

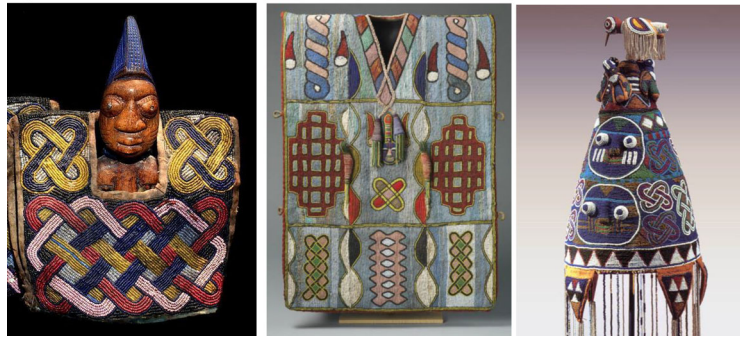


Figure 3

Interlaces in Yoruba sacred art (Nigeria). At left, Ibeji statue, courtesy Lenard Kahan gallery. At center, a tunic for ritual occasions, courtesy of Princeton University Art Museum. At right, a royal crown (*adènlà*), courtesy of the Annenberg Foundation.

Yoruba Interlace Patterns

In the areas spanning Nigeria to Benin, practices that focus on Orishas – entities that mediate between humans and the divine – are manifested in a variety of spiritual practices. They are especially important in the context of Africa’s cultural contributions in the Americas because of their presence in diasporic religions. For example, in the Candomblé practice in Brazil, they are referred to as “*orixás*.” They also appear in Cuban Santería, Haitian Vodou, and related practices under various names and associations.

Figure 3 shows interlace patterns in beadwork from Yoruba sacred art associated with the Orisha deities in Nigeria. Because the patterns are complex, yet form a closed loop (“unicursal”), they are often interpreted as symbolizing a spiritual power, *ase*, that winds through everything, and yet is never-ending. “Interlaces on beaded panels proclaim that past, present, and future are part of the infinite cycle of existence: birth, life, departure, return” (Drewal and Mason 1998, 18). Sometimes that is more specifically described in relation to ancestors. For example, Annenberg (2009) describes interlace on the Yoruba ceremonial crown (*adènlà*) that is referred to as “the chain of divine ancestors and continuity of life.” In other contexts the interlace patterns represent related aspects of vitality and life force. For example, interlace patterns in both 2D and 3D form appear in the masks of Yoruba Gelede ceremonies (Figure 4). Drewal (1979) describes how these mask designs often imitate women’s textile head wraps and braids.

Through these and other motifs, “performers externalize the vital natures of men and women, projecting an image of vital life force, *ase*” in the Gelede masks, costumes, and dance (Drewal 1979, 227).

Thus across many aspects of Yoruba art, the interlace forms are associated with a concept of spiritual energy and its circulation of the vital life force. As noted previously, this is a close match to the concept of generative justice. Eglash (2016) explains the concept with



Figure 4

Interlaces in headwrap and braiding depictions in Gelede ceremony masks. At left: courtesy of the Smithsonian. Center: courtesy of Babatunde Lawal. At right: courtesy of Newfields Museum.

the example of organic agriculture. Industrial agriculture extracts value from soil, causing nutrient depletion. In organic agriculture, waste is composted and returned to the soil, maintaining the regenerative capacity of the soil organisms. This exemplifies the regeneration of ecological value. In the same way, the extraction of labor value, as Marx phrased it, exhausts workers' ability to regenerate. Worker-owned business, cooperatives, and collectives (a common application of "ubuntu") and similar structures can return labor value to workers, creating less alienated work. And social value can be extracted: for example, social media colonizes our human networks of friends, family, and broader communication. Online systems like Wikipedia and physical systems like community-based development allow society to regenerate its own forms of value. In sum: generative justice (which was originally inspired by Indigenous systems) specifies that ecological value, labor value, and social value should all circulate from the source of generation back to that source.

One challenge for humanity is that many aspects of our psychology – greed, selfishness, fear; blind obedience to systems of authority or ownership – can break those cycles of regeneration through value extraction, creating forms of alienation (e.g. consumerist society) that trap us into systems of exploitation. African interlace patterns operate as a kind of visualization technology to keep concepts like mutual aid, shared life force, and so on at the center of the community's conceptual and material practice, and opposing our worst instincts and social tendencies.

Other Examples Across African Geographies and Time Periods

What are we to make of the great variety of interlace practices throughout the African continent, and deep into its history? There are interlace patterns in Kongo ivory horns from the sixteenth century, in the "woven ware" ceramics of Central Africa dating back to the twelfth century, in the carved-rock Ethiopian churches dating back to the eleventh century, in the ninth-century bronze of the middle Niger, in the sixth-century ceramics of the Cross River area of Nigeria, and

perhaps even older examples (Cranshof et al. 2018; Lagamma 2021; Slogar 2007). That variety has some correspondence to the vastly different ways of preventing value alienation, and ensuring instead value circulation, across African traditions.

For example, the Ghanaian adinkra symbol *Mpatapo*, “the knot of reconciliation,” is a very direct, literal-minded version of this correspondence. The Yoruba visualizations of *ase*, the spiritual life force, are more abstract and indirect. The abundant metaphors for knotting associated with Lusona-practicing cultures, as noted above, are somewhere in-between. The concept of *ubuntu* or “mutual aid” appears in a wide variety of forms, iconographies, and media in Africa: one could even make a case for the famous “call and response” music pattern as acoustic interlace, and indeed the connection between African music practices and mutual aid is well established (Rudge 2019). In other words, if we look at the total set of Indigenous systems for generative justice, and the total set of Indigenous visualizations for interlace, we will see not a reduction to single meaning, but rather expansion within families of resemblance. We will now turn to these families of resemblance in the interlace forms created by African descendants in the Americas, and examine the generative forms of resistance, flourishing, and innovation that emerged from those “heritage algorithms” (Bennett 2016).

Interlace in the African Diaspora

Interlace in African American Braiding

The connection to braiding and headwrap patterns for women in the Gelede designs from Figure 4 is especially poignant in the history of American slavery, where the excuse of shaving heads for sanitary reasons was part of the assault on personhood. Since that time the recovery of Black hairstyle tradition has been a flashpoint for resistance and empowerment, from slavery to reconstruction to Jim Crow to civil rights to contemporary labor rights for professional braiders codified in the CROWN act⁶ (Donahoo and Smith 2022; Johnson 2016). The interlace patterns in contemporary African American braiding (Figure 5) are thus both contemporary innovations and yet simultaneously bridges to deep forms of cultural, artistic, spiritual, and mathematical heritage.

Swept Yard Patterns in African American Gardens

African American gardening traditions in the American south include areas in which sweeping maintains bare earth. Observations of this “swept yard” tradition date back to 1725, and they are often cited in relation to similar traditions in West Africa (Fesler 2010). Brown (1999) quotes one of her informants (born in 1930), who described his grandmother’s insistence on creating specific patterns in the swept yard earth. Joachim (1999, 74) interviewed African Americans



Figure 5

Interlace designs in contemporary African American braiding.

in Texas, and had similar reports from two of her informants: grandmother “would have little circles and stuff” in the swept earth. While no one mentioned a specific lusona style of continuous line drawing, the parallels are intriguing. The most famous case of pattern drawing in African American swept yards is from Maya Angelou’s *I know why the caged bird sings*. She describes the designs they drew in the bare earth yard, their destruction during a potentially dangerous encounter with the white girls, and the design’s regeneration with a recursive pattern:

I took the rake back to the front yard. The smudged footprints were easy to erase. I worked for a long time on my new design and laid the rake behind the wash pot. When I came back in the Store, I took Momma’s hand and we both walked outside to look at the pattern. It was a large heart with lots of hearts growing smaller inside, and piercing from the outside rim to the smallest heart was an arrow. Momma said, ‘Sister, that’s right pretty’.
(Angelou 1969, 26)

Haitian Vèvè as Dynamic Grids

Figure 6 is a vèvè diagram from the Caribbean island of Haiti. Not all vèvè correspond well to interlace patterns, but many have a grid-like basis. We can immediately see the similarity to both the Yoruba interlace patterns from Nigeria and the lusona interlace patterns from Angola. Rather than etched into earth, they are drawn by pouring powder. The powder is typically a mix of cornmeal and ash, symbolizing the generative cycling between life and death.

Note that in the video still image of Figure 6, the right-most column of X shapes has been filled, and the priest is now starting on the next column over. Each X is created with one scoop of the powder. When finished, the priest reaches for another scoop. This rhythmic drawing is often accompanied by music, emphasizing the linkage of processes in space and time.

Unlike lusona, the emphasis is not on a single continuous line. But, like lusona, the movement, trajectory, and rhythm is as important as the finished artifact. Benjamin Hebblethwaite, noting the many different spirits or “lwa” in Vodou, describes this improvisation in his description of the video of the ritual practice above:

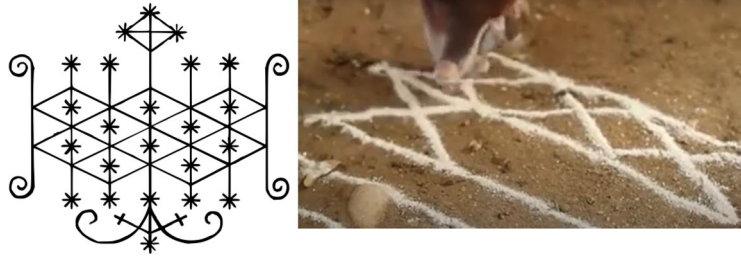


Figure 6

Left: Vèvè for Ogun. Right: still photo from a drawing ritual video, showing another grid-like vèvè, held in Croix-des-Bouquets, Haiti, August 2013. By permission of Benjamin Hebblethwaite and the Archive of Haitian Religion and Culture at <https://dloc.com/collections/vodou>

The vèvè are partially improvisational but they also demonstrate a traditional set of patterns that correspond to the many different Vodou lwa as well as a common aesthetic vocabulary of embellished decoration characterized by ornate stars and filigree swirls. While there are certain canonical images that correspond to the various lwa, there exists no definitive or exhaustive list of the veve. (Hebblethwaite 2016)

In sum: despite the switch from unicursal paths etched in earth, to rhythmic, iterative powder depositions, vèvè are as much a part of the dynamic grid family of practices as the others we highlight here.

Capoeira in Brazil

African interlace patterns also appear in Brazilian iconography. One of the most interesting possibilities for this connection is in the martial arts performance of capoeira. According to tradition, capoeira began as a means of disguising practice for slave rebellion, under the guise of dance. It was first mentioned in colonial documents under the name *capoeiragem* in 1789, as “the gravest of crimes.” But it also sustained spiritual and cultural aspects of African heritage. Oral tradition describes an origin in Angola, and indeed a similar fusion of dance and martial arts can be found there, called *N’golo*. Angola is also the central location for lusona practices. Dossar (1992) notes the connection by describing how both capoeira in Brazil and *N’golo* in Angola incorporate music, rhythm, and drawing on the ground prior to the performance initiation (*saida*).

But the primary interlace structure in both Brazilian capoeira and Angolan *N’golo* is not the 2D sand path. Rather it is the 3D body movement path, in which two or more performers create acrobatic intertwinings that are constantly on the verge of collision, yet manage to route around points of potential intersection. Annika Kreikenbohm (2017), an astrophysicist and information designer, captured some of capoeira’s bodily interlace patterns by strapping LED lights to the performers, and using long exposure photography (Figure 7).

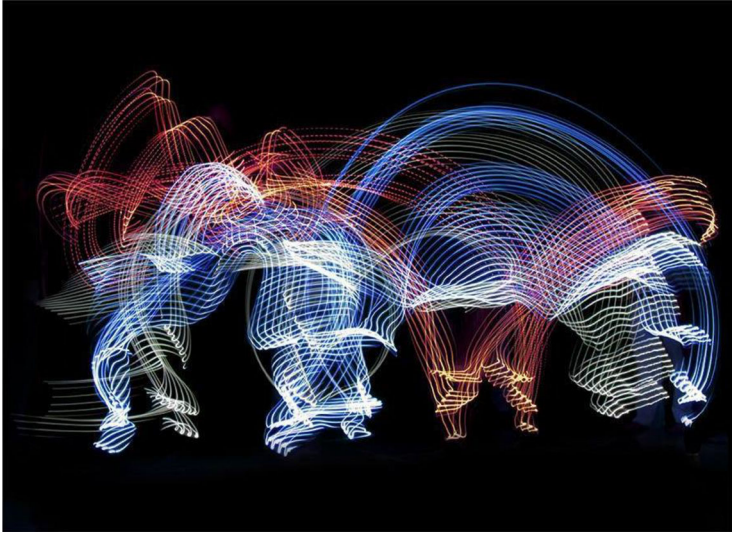


Figure 7

Long exposure photography used to trace capoeira's movement interlace paths. Image copyright Annika Kreikenbohm.

African Interlace in Diaspora Design Ecologies: Weaving Together Formal and Social Dimensions

The contrast between the formal aspects of African interlace – the Indigenous knowledge system composed of these heritage algorithms in cultural context – and their Western mathematics and computation counterparts is quite striking. In the West, the purpose of discovering underlying properties is often to create a singular universal that can now be dislocated, what Latour (1990) called “the immutable mobile” of science. The dislocation, in the context of exploitative practices, allows it to be exclusively secured by intellectual property rights, to create metrics for optimizing the extraction of value from workers and nature, to police conceptual and material boundaries that stray from its confines, and so on.

There are occasionally Western instances, such as open source, which defy this trend, and focus instead on universals in the service of generative justice. Conversely, there are occasionally African instances in which an interlace is exclusively reserved as the royal style (Schildkrout and Keim 1990), just as Europeans would reserve IP rights. The reason the exceptions exist is because there are exceptions to the general trend toward capitalist economies in the West and ubuntu economies in Africa. The exceptions for Western societies are the relatively rare moments when nonhierarchical networks allowed generative practices, as we saw with older forms of maintenance of the village commons, newer hacker networks that developed open source, and so on. The exceptions for traditional African societies are relatively rare moments when wealth

accumulation allows Western-style class stratification, as we saw with the kingdom of Benin.

But in the vast majority of traditional African societies, abstract principles symbolized by geometries invoking cycles of exchange, recursion, symmetry, mutuality, and other properties closely linked to “anti-extraction” were common decorative motifs. We have seen this with the Isona cultures’ metaphors of knots, Yoruba’s *ase*, the Ghanaian *mpatapo* symbol, and other examples. The “unicursal” paths that end where they began are reminders about the social conventions that ensure that value is circulated rather than alienated (Eglash 2016; Gowdy 1999; Ignatov 2016).

We propose a general term for bringing together Indigenous formalisms in the context of designs for more just and sustainable forms: “diaspora design ecologies.” The terminology was inspired by Imani Jacqueline Brown’s (2023) “Black Ecologies” anthology, which featured artists and designers examining the intersections of Indigenous practices, modes of resistance, and environmental resonance, in particular in relation to African fractals. Related approaches might include the emerging synthesis between African American land traditions and regenerative farming in the United States (Layman and Civita 2022), as well as more abstract frameworks such as ethno-biomathematics (Eglash 2023). Diaspora design ecologies thus encompass the application of cultural arts, Indigenous regenerative traditions, and their corresponding formalisms (computational, architectural, systemic, etc.) to design practices for all environments, both built and natural, from urban to rural, and design at all scales, from micro products to global economies, using African principles for egalitarian, democratic, and sustainable lifeways.

One challenge in diaspora design ecologies is negotiating a path between acceptance and resistance. For example, “acceptance” is often embedded in the rhetoric around Marvel’s first Black Panther movie, which earned US\$1.3 billion in the first month. That is not necessarily a bad thing: we should rightly celebrate the genius of Ruth Carter, costume designer; Chadwick Boseman, the late actor; and others in their accomplishments. African interlace played a role for both: consider, for example, the transition from traditional interlace on the shirt in Figure 8 left, and its creative reinterpretation on the suit at right. The Wakanda salute in that image is also a form of interlace at its most basic element, the over-under crossing of arms that envision flow back onto the body itself. According to Stewart (2021, 60), director Ryan Coogler has stated that the crossed-arms gesture stems in part from West African sculptural forms.⁷

On the other hand, very little of the US\$1.3 billion in wealth was transferred to Black communities: in that aspect, acceptance for the few means appropriation of cultural capital from the many. If resistance had taken the form of a boycott – a refusal to participate in the Black Panther movie production – it would have done little to stop it.



Figure 8

Interlace in Black Panther's costume design: traditional at left, reinterpretation at right.

Fortunately, we need not choose between these unacceptable outcomes: we can weave a developmental trajectory between them.

Interlace structures offer conceptual tools that avoid the forced choice between acceptance for the few or futile resistance by the many. They offer an exciting opportunity to explore the transformative power of diaspora design ecologies: a means by which the practices and eco-epistemologies on the Indigenous side, which emerged in the service of ubuntu's egalitarian economies and traditional sustainability practices, can be interwoven with the power of Western formalisms. Applications might include tools for monitoring, modifying, and sustaining value flows, such that they develop into egalitarian and sustainable economies. These tools could include interlace-inspired visualizations to enhance intuition and understanding, or topological metrics (e.g. the scaling exponent of degree distributions of value chain networks) for quantitative analysis. In the literature on generative justice, we often refer to circular flows for ecological value, labor value, and social value. It is useful to separate them as an analytic tool, but the design of real-world egalitarian and sustainable community-based economies will require ways of creating braided, intertwined, or other interlaced flows between these three domains (Eglash et al. 2021, 2024).

Less abstract, more practice-based interpretations might include biomaterial structures that draw mechanical strength from interlaced fibers, thus retaining ecological sustainability in low-carbon production and compostable products, or new designs for carbon fiber mesh that enhance strength, or new architectural layouts and urban design forms that better serve the needs of citizens. This is contrary to the trend for industrial mass production, in which the complex fibers of biological entities are flaws that must be reduced to uniform substances. Such industries are essentially eating everything, and shitting out unsustainable, toxic plastics. But new technologies such

as AI can now approach natural states of substances less as unpredictable flaws and more as information-rich interlaced structures. This offers a basis for interweaving biological processes into living architectures, and limiting products and processes to those that are unalienated partners in unicursal ecological flows.

Black Interlace Scholars of Today

Bennett’s “agentic design” in visual process for education, Felecia Davis’ “dreadlock series” in architecture, and Kwame Robinson’s “solidarity routing” for worker-owned e-delivery are all examples in which contemporary interlace concepts arise at the intersection of computation, cultural tradition, and diaspora design ecologies.

Bennett, a professor of art and design at Stamps school, University of Michigan, describes her work involving computational models for STEM education, GUI design in human-computer interaction, and other domains as “agentic design” (Bennett 2021; 2023). She notes how this approach allows the process of emergence, organically developing from the bottom-up, rather than imposing a “master designer’s vision” from the top-down. Like the call-and-response acoustic interlace, the designer is creating a context in which the user is neither abandoned on their own, nor forced into restricted behaviors, but rather guided as they co-create a design trajectory. Both the original African interlace and their diaspora cousins are among the “heritage algorithms” she draws from. Figure 9 shows some of the STEM education processes she utilized in her development of CSDTs (“Culturally Situated Design Tools”) to allow



Figure 9
Bennett’s agentic design approach allows students to explore computational models for interlace heritage algorithms ranging from hexastrip weaving, to cornrows, to quilting, and expressed both virtually and as physical renders.

students computational explorations of traditional and vernacular interlace traditions, ranging from rattan hexastrip weavings to cornrows to textiles. She summarizes the goals as follows:

The outcomes of these workshops and students' interaction with CSDTs are aimed not at tricking children into the STEM pipeline, but at transforming STEM itself, potentially creating new kinds of scientists and engineers who no longer see their discipline as inherently divorced from cultural, economic and ecological dimensions of the world. [... Thus] the end goal is not limited to reducing the inequality in rates of STEM participation, but also to strengthen the flow of unalienated value between schools and communities, younger and older generations, and between the disciplines themselves. (Bennett 2021, 12).

Felecia Davis, an associate professor of architecture at Penn State, refers to her area of research as “networked architectural textiles.” In the “Dreadlock Series” she describes her process as follows:

‘These started out as flat textiles using what we call African fractals or a Fibonacci series to make the rose,’ explained Davis. ‘I then suggested that these roses could be popped up to be used as tension structures or with a bio-based resin to make an architectural structure that can be shaped using gravity or tension’.⁸

Structural responses to tension are just one way in which Davis' dynamic grids respond to their environments. In [Figure 10](#) at right,



Figure 10
Responsive textile-based architectures by Felecia Davis, professor of architecture at Penn State. Images courtesy of Felecia Davis.

we see one of her heat-sensitive fabrics that create their own holes, creating models for architectures that adapt to shifting ground, wind patterns, or other environmental conditions. At left is a photosensitive structure that shifts its shape in response to changes in lighting. Davis' wide range of dynamic grids encompass everything from mood sensing (Davis et al. 2013) to touch (Davis 2018) to structural tension (Oghazian et al. 2023). Her "Black Reconstruction Collective" (<https://www.blackreconstructioncollective.org/>), a network of Black architects, artists, designers, and scholars whose mission is "multi-scalar and multi-disciplinary work dedicated to dismantling systemic white supremacy and hegemonic whiteness within art, design, and academia," offers one possible example of what we are describing as diaspora design ecologies.

Kwame Robinson, an assistant professor at Wayne State University in Detroit, investigates how worker-owned computational facilities can develop e-delivery routing that better empowers their work-life experiences and financial gains (Robinson 2024). Interlace forms naturally emerge from this work because the routes are not optimized in the way they would be in DoorDash or Uber.

Robinson begins with images of the grid imposed by settler-colonial farms in 1875, which sought to maximize extractive value flows to river transportation, and their conversion to the contemporary Detroit street grid. His e-delivery alternative, "Solidarity Pathways," then asks what kind of dynamic routing would weave through this grid in ways that minimize value extraction. Urban farms in Detroit have long-standing ties to local social justice activism, and thus made a strong choice for one set of nodes for this emergent network. He partnered with Deeply Rooted Produce, which sources from BIPOC-owned (Black, Indigenous, and People of Color) and -operated family farms, as well as their own locally grown produce. Deeply Rooted processes the harvests and delivers to both relevant market channels in Metro-Detroit as well as directly to customers.

By coding their own e-delivery app, Robinson's team was able to test different design configurations, including both driver and supplier experiences. One set of extractive flows came from the per-transaction fees that the current routing, delivery, and payment software created. Others were more subtle aspects of what is absent: since software is generally attempting to optimize for corporate profits, issues such as driver contingencies (what time I need to pick up my kids, routing for safety rather than speed, and so on) are made invisible. The resulting interlace of routing is both dynamic and generative, adapting to local needs and expanding the possibilities for community-based economic solidarity.

Concluding Remarks and Future Research

As noted by Pais (2011), the temptation in ethnocomputing, ethnomathematics, and similar "translations" of Indigenous knowledge into formal structures is the mode of acceptance, as if to say: "You see?

They are not primitive, because their math is just like your math.” But in doing so, we lose the unique understandings that are possible, including modes of resistance, on both Western and non-Western sides of that translation. African interlace enjoys participation in the family of formalisms related to topology and knot theory, but they offer new insights as well, related in part to their links to generative systems. For example, while a single instance of a unicursal loop symbolizes a kind of regenerative system (e.g. life/birth), the systems for generating recursive loops – the age-grade advancement of wisdom, the intergenerational accumulations that create long-term forms of environmental and social meta-adaptation (e.g. Dogon epistemology as described in Eglash 1999, 131–3) – act as a kind of meta-algorithm for the regeneration of regenerative loops. It is the kind of meta-recursion that Cantor formalized as transfinite set theory, Gödel as his incompleteness theorem, and Turing as computability.⁹ But whereas the Western approach tends to focus on defining and policing completeness, the African tradition trends toward a focus on flourishing.

This distinction is increasingly important as the Western approach has increasingly switched from domination through top-down control to domination through the *appropriation* of bottom-up systems. Perhaps nothing exemplifies this more than the phrase “generative AI.” After decades and billions of funding, attempting AI as top-down systems of explicitly programed rules and predefined categories failed. The bottom-up approach of generative AI collects statistical patterns of images and text appropriated from the Internet, without returns of value to those who created that data. Social media also exemplifies this trend: rather than control from the top-down, as portrayed in something like Orwell’s *1984*, social media colonizes our own social networks from the bottom-up, turning what should be social regeneration of democratic and grassroots commitments into new forms of commodification and exploitation.

African interlace provides an alternative model: bottom-up organization dedicated to its original purpose of generative justice. It conveys this by coupling the vivid imagery of recursive pathways with rich cultural meanings: the circulation of “life force” described by “nummo” (Dogon), “ase” (Yoruba), “mooy” (Luunda), and so on. As we saw with Robinson’s “Solidarity Pathways,” Bennett’s design tools, and Davis’s computational architecture, it is possible to redesign AI and related technologies to better conform with this original purpose of interlace forms as a means to return ecological, labor, and social value back to its original generative source.

As we explore Black interlace as heritage algorithms, new possibilities for their multidisciplinary applications continue to emerge. Simply engaging in further exploration of their origins in Africa can be fruitful. In the same way, their implications as “Africanisms” in the Americas offers a variety of new investigatory paths for history, archaeology, cultural anthropology, and Black studies. And by further

understanding them as heritage algorithms – traditional computational formalisms that bridge the gap between concrete practice and abstracted meaning – they open up new pathways for diaspora design ecologies, and new means of merging Western computational technologies with the traditional African emphasis on forms of generative justice.

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Notes

1. Steil and Carvalho (2014) date the use of this concept from Indigenous origins, to its re-interpretation and translation to Western terminology by anthropologists such as Batson and Ingold, to STS and New Materialism scholars.
2. Ubuntu philosophy roughly translates to the European concept of “mutual aid.” But Western frameworks of game theory, where mutual aid is maximizing self-fitness, is a poor match to ubuntu philosophy, which is often summarized as “I am because we are”; that the “self” is social and not individual (Gade 2011).
3. In 1736, Euler introduced graph theory with a story about his city of Königsberg: local myth claimed you could traverse all seven of its bridges without visiting any twice, but he showed that such a “unicursal” path is not possible. Graph theory terminology still reflects the concept of paths. A “walk” refers to connecting lines at vertices, and a “trail” is a walk in which no line is visited twice. A trail that ends where it began is “unicursal.” If the trail includes the concept of over/under distinctions, we have knot theory. If a trail is inscribed on a specific surface, such as sphere versus donut, we have topological graph theory.
4. An easy way to facilitate this is to imagine that the grid is in a box of mirrors, and the line is a laser beam. Starting with a beam at the diagonal between dots, and knowing that angle of incidence = angle of reflection, you can complete the trace for any size grid in the $(n + 1)$ by $(n + 2)$ family.
5. Eglash (1999, 131–3) describes this in detail: the meta-algorithm in Dogon epistemology is itself embedded within larger regenerative loops. That is to say, the algorithms for one set of Dogon patterns are themselves created by higher-level algorithms. These meta-levels represent Dogon cosmology as the continual unfolding of a regenerative universe.
6. The CROWN Act stands for “Creating a Respectful and Open World for Natural Hair” and is a legislative act that aims to prohibit discrimination based on a person’s hair texture or hairstyle, essentially protecting people from being discriminated against for wearing natural hairstyles commonly associated with a particular race or ethnicity, like cornrow braids, locs, or twists.
7. For images of these sculptures, and traditional Africa use of the crossed-arm gesture, see De Grunne (1988).
8. Quoted from <https://arts.psu.edu/news/architecture-labs-computational-textiles-work-featured-in-upcoming-exhibition/>
9. Cantor erased the middle of a line, then erased the middle of the two remaining lines, and repeated this to infinity, creating the first fractal in Western mathematics. What Cantor did for geometry, Gödel did for set theory, and Turing did for computing, in each case turning the system back on itself to show the limits of understanding when recursion is applied.

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References

- Angelou, Maya. 1969. *I Know Why the Caged Bird Sings*. NY: Random House.
- Annenberg. 2009. "Art: Royal Crown (adènà)." *Art Through Time: A Global View*. <https://www.learner.org/series/art-through-time-a-global-view/ceremony-and-society/royal-crown-adenla/>.
- Babbitt, Bruce, David Lyles, and Ron Eglash. 2012. "From Ethnomathematics to Ethnocomputing: Indigenous Algorithms in Traditional Context and Contemporary Simulation." In *Alternative Forms of Knowing (in) Mathematics*, 205–219. Leiden: Brill.
- Bennett, Audrey G. 2003. "Towards an autochthonic black aesthetic for graphic design pedagogy." *Journal of Design Research* 3 (2): 61–70.
- Bennett, Audrey G. 2016. "Ethnocomputational Creativity in STEAM Education: A Cultural Framework for Generative Justice." *Teknokultura. Revista de Cultura Digital y Movimientos Sociales* 13 (2): 587–612. https://doi.org/10.5209/rev_TEKN.2016.v13.n2.52843.
- Bennett, Audrey G. 2021. "Agentic Design: An Emergent Approach to Generative Justice." *New Design Ideas* 5 (1): 5–20.
- Bennett, Audrey G. 2023. "Heritage Algorithms Combine the Rigours of Science with the Infinite Possibilities of Art and Design." *The Conversation*. <https://theconversation.com/heritage-algorithms-combine-the-rigors-of-science-with-the-infinite-possibilities-of-art-and-design-209309>.
- Berry, Anne H., Kareem Collie, Penina Laker, Lesley-Ann Noel, Jennifer Rittner, and Kelly Walters, eds. 2022. *The Black Experience in Design: Identity, Expression and Reflection*. Simon and Schuster.
- Brown, DeNeen L. 1999. "Black Hands, Green Gardens: In African American Yards, Nurturing the Roots." *The Washington Post*, August 25, 1999. <https://www.washingtonpost.com/archive/life-style/1999/08/25/black-handsgreen-gardens/a000b81a-ad14-4da9-ab5b-7be443e266a4/>.
- Cranshof, Els, Nsiami Nikis, and Pierre de Maret. 2018. "Ceramics Decorated with Woven Motifs: An Archaeological Kongo Kingdom Identifier?." In *The Kongo Kingdom: The Origins, Dynamics and Cosmopolitan Culture of an African Polity*, edited by Koen Bostoen and Inge Brinkman, 165–196. Cambridge: Cambridge University Press.

- Davis, Felecia. 2018. "Touch: Communication of Emotion Through Computational Textile Expression." In *Distributed, Ambient and Pervasive Interactions: Technologies and Contexts—6th International Conference, DAPI 2018, Held as Part of HCI International 2018, Las Vegas, NV, USA, July 15–20, 2018, Proceedings, Part II*, edited by [Editor(s) if known], 292–309. Lecture Notes in Computer Science 10725. Cham, Switzerland: Springer International Publishing.
- Davis, Felecia, Asta Roseway, Erin Carroll, and Mary Czerwinski. 2013. "Actuating Mood: Design of the Textile Mirror." In *Proceedings of the 7th International Conference on Tangible, Embedded and Embodied Interaction (TEI 2013)*, 99–106. Barcelona, Spain: Association for Computing Machinery (ACM).
- De Boeck, Filip. 1993. "Symbolic and Diachronic Study of Intercultural Therapeutic Divinatory Roles Among Aluund ("Lunda") and Chokwe in the Upper Kwango (Southwest Zaire)." *Afrika Focus* 9 (1-2): 73–104. <https://doi.org/10.1163/2031356X-0090102005>.
- De Grunne, Bernard. 1988. "Ancient Sculpture of the Inland Niger Delta and Its Influence on Dogon Art." *African Arts* 21 (4): 50–92. <https://doi.org/10.2307/3336745>.
- Donahoo, Saran, and Asia D. Smith. 2022. "Controlling the Crown: Legal Efforts to Professionalize Black Hair." *Race and Justice* 12 (1): 182–203. <https://doi.org/10.1177/2153368719888264>.
- Dossar, Kenneth. 1992. "Capoeira Angola: Dancing Between Two Worlds." *Afro-Hispanic Review* 11 (1/3): 5–10.
- Drewal, Henry John. 1979. "Pageantry and Power in Yoruba Costuming." In *The Fabrics of Culture: The Anthropology of Clothing and Adornment*, edited by Justine M. Cordwell and Ronald A. Schwarz, 189–230. The Hague: Mouton Publishers.
- Drewal, Henry John, and John Mason. 1998. "Beads, Body, and Soul: Art and Light in the Yoruba Universe." *African Arts* 32 (3): 1.
- Eglash, Ron. 1999. *African Fractals: Modern Computing and Indigenous Design*. Rutgers, New Jersey.
- Eglash, Ron. 2016. "Of Marx and Makers: An Historical Perspective on Generative Justice." *Teknokultura. Revista de Cultura Digital y Movimientos Sociales* 13 (1): 245–269. https://doi.org/10.5209/rev_TK.2016.v13.n1.52096.
- Eglash, Ron. 2023. "Ethno-Biomathematics: A Decolonial Approach to Mathematics at the Intersection of Human and Nonhuman Design." In *Ubiratan D'Ambrosio and Mathematics Education: Trajectory, Legacy and Future*, edited by Marcelo C. Borba and Daniel C. Orey, 289–303. Cham: Springer International Publishing.
- Eglash, Ron, Audrey Bennett, Michael Lachney, and William Babbitt. 2021. "Evolving Systems for Generative Justice: Decolonial Approaches to the Cosmo-Local." In *The Cosmo-Local Reader*, edited by José Ramos, Sharon Ede, Michel Bauwens, and James Gien Wong, 60–75. Melbourne: Futures Lab.

- Eglash, Ron, Kwame P. Robinson, Audrey Bennett, Lionel Robert, and Mathew Garvin. 2024. "Computational Reparations as Generative Justice: Decolonial Transitions to Unalienated Circular Value Flow." *Big Data & Society* 11 (1): 20539517231221732. <https://doi.org/10.1177/20539517231221732>.
- Fesler, Garrett. 2010. "Excavating the Spaces and Interpreting the Places of Enslaved Africans and Their Descendants." In *Cabin, Quarter, Plantation: Architecture and Landscapes of North American Slavery*, edited by Clifton Ellis and Rebecca Ginsburg, 27–49. New Haven, CT: Yale University Press.
- Gates, Henry Louis, Jr. 2014. *The Signifying Monkey: A Theory of African American Literary Criticism*. Oxford: Oxford University Press.
- Gerdes, Paulus. 1994. "On Mathematics in the History of Sub-Saharan Africa." *Historia Mathematica* 21 (3): 345–376. <https://doi.org/10.1006/hmat.1994.1029>.
- Gowdy, John. 1999. "Hunter-Gatherers and the Mythology of the Market." In *The Cambridge Encyclopedia of Hunters and Gatherers*, edited by Richard B. Lee and Richard Daly, 391–398. Cambridge: Cambridge University Press.
- Griaule, Marcel. 1965. *Conversations with Ogotemmelé*. London: Oxford University Press.
- Hebblethwaite, Benjamin. 2016. "Kwadèboukè Seremoni – Veves (Part 1)." Youtube video, 5:59. Posted April 26, 2016. <https://www.youtube.com/watch?v=bdxHuXCLIdc>.
- Ignatov, Anatoli. 2016. "African Orature as Ecophilosophy: Tuning in to the Voices of the Land." *GeoHumanities* 2 (1): 76–91. <https://doi.org/10.1080/2373566X.2016.1166977>.
- Joachim, Sally A. 1999. "African-American Garden Culture: A Study of Tradition and Translation." PhD diss., Louisiana State University. https://repository.lsu.edu/gradschool_disstheses/8296.
- Johnson, Elizabeth. 2016. *Resistance and Empowerment in Black Women's Hair Styling*. London: Routledge.
- Kreikenbohm, Annika. 2017. "EMotion—The Energy Flow in Capoeira." *Annok*. <https://annok.de/ledcapoeira/>.
- Kubik, Gerhard. 1987. "African Space/Time Concepts and the 'Tusona' Ideographs in Luchazi Culture." *African Music: Journal of the International Library of African Music* 6 (4): 53–89.
- Lagamma, Alisa. 2021. *Sahel: Art and Empires on the Shores of the Sahara*. New York: The Metropolitan Museum of Art.
- Latour, Bruno. 1990. "Drawing Things Together." In *Representation in Scientific Practice*, edited by Michael Lynch and Steve Woolgar, 19–68. Cambridge, MA: MIT Press.
- Layman, Emma, and Nicole Civita. 2022. "Decolonizing Agriculture in the United States: Centering the Knowledges of Women and People of Color to Support Relational Farming Practices." *Agriculture and Human Values* 39 (3): 965–978. <https://doi.org/10.1007/s10460-022-10297-3>. [35106023]

- Oghazian, Farzaneh, Sam Moradzadeh, and Felecia Davis. 2023. "Form-Finding of Architectural Knitted Tensioned Structures." In *42nd Annual Conference of the Association for Computer Aided Design in Architecture*, edited by Masoud Akbarzadeh, Hina Jamelle, and Robert Stuart-Smith, 326–333. Fargo: Acadia Publishing Company.
- Pais, Alexandre. 2011. "Criticisms and Contradictions of Ethnomathematics." *Educational Studies in Mathematics* 76 (2): 209–230. <https://doi.org/10.1007/s10649-010-9289-7>.
- Robinson, Kwame. 2024. "Computing for Community-Based Economies." PhD diss., University of Michigan.
- Rudge, Alice. 2019. "Flexibility and Egalitarianism: Musical Insights from Hunter-Gatherers." *Ethnomusicology Forum* 28 (2): 163–183. <https://doi.org/10.1080/17411912.2019.1683875>.
- Schildkrout, Enid, and Curtis A. Keim. 1990. *African Reflections: Art from Northeastern Zaire*. Seattle: University of Washington Press.
- Slogar, Christopher. 2007. "Early ceramics from Calabar, Nigeria: towards a history of nsibidi." *African Arts* 40 (1): 18.
- Steil, Carlos Alberto, and Isabel Cristina de Moura Carvalho. 2014. "Epistemologias Ecológicas: Delimitando um Conceito." *Mana* 20 (1): 163–183. <https://doi.org/10.1590/S0104-93132014000100006>.
- Stewart, Felicia. 2021. "Cross My Heart and Hope to Die in Wakanda: Expressions of Solidarity in Black Panther." In *Why Wakanda Matters: What Black Panther Reveals About Psychology, Identity, and Communication*, edited by Sheena C. Howard, 3–16. Dallas: BenBella Books.